

Explore the network

Start here

In WSL: choose an exercise. Each command starts it afresh.

```
make networking      # follow a request
make operations      # find the fault
make automation      # repair through an API
make monitoring      # measure traffic and loss
```

Help in WSL: make task, make hint STEP=2, make solution STEP=2.

Enter the Lagoon probe:

```
make enter NODE=probe01.bob1.lagoontransit.test
```

For the other probe, replace lagoontransit with pacifictransit. Leave Linux with exit, a router CLI with quit.

Inside a probe

```
ip -br addr          # my addresses
ip route get 198.51.100.10 # gateway and source
dig @198.51.100.10 data.oceanresearch.test A +time=1 +tries=1
ping -4 -c 3 -W 1 198.51.100.10 # IPv4 reachability
ping -6 -c 3 -W 1 2001:db8:100::10 # IPv6 reachability
curl --fail --max-time 5 http://data.oceanresearch.test/
```

Look inside a router

In WSL: enter ReefNet Edge 01.

```
make enter NODE=edge01.bob1.reefnet.test
```

For Lagoon, use NODE=edge01.bob1.lagoontransit.test.

SR Linux CLI

Interfaces and routing sessions:

```
show interface brief
show network-instance default protocols bgp neighbor
show network-instance default protocols ospf neighbor
```

Customer prefix: BGP and active route table.

```
show network-instance default protocols bgp routes ipv4 prefix 198.51.100.0/24
show network-instance default route-table ipv4-unicast prefix 198.51.100.0/24
```

On ReefNet Edge 01, what do we advertise to Lagoon? Which policies apply?

```
show network-instance default protocols bgp neighbor 192.0.2.2 advertised-routes ipv4
info from running network-instance default protocols bgp
info from running routing-policy
```

Read and change configuration

Read through an API

In **WSL**: enter the operator workstation.

```
make enter NODE=operations01.bob1.reefnet.test
```

Run the commands below **inside operations01**.

Find a device in NetBox

```
TOKEN=$(cat /state/netbox-token)
NETBOX_API=http://netbox.bob1.reefnet.test:8080/api
curl -fsSG -H "Authorization: Bearer $TOKEN" \
  "$NETBOX_API/dcim/devices/" \
  --data-urlencode name=edge01.bob1.reefnet.test |
jq '{count, devices: [.results[] | {name, primary_ip4}]}'
```

Expect one device. Check its name and primary IP. Keep the token private.

Read the BGP configuration with gNMI

```
BGP_PATH='/network-instance[name=default]/protocols/bgp'
gnmic -a edge01.bob1.reefnet.test:57400 \
  -u admin -p 'NokiaSr11!' --skip-verify --encoding json_ietf \
  get --type config --path "$BGP_PATH"
```

Change a BGP policy with gNMI

Still inside operations01: read the affected group and note its current policy.

```
BGP_PATH='/network-instance[name=default]/protocols/bgp'
GROUP="$BGP_PATH/group[group-name=lagoon-v4]"
gnmic -a edge01.bob1.reefnet.test:57400 \
  -u admin -p 'NokiaSr11!' --skip-verify --encoding json_ietf \
  get --type config --path "$GROUP"
```

Remove the group's policy override

After diagnosis: the group uses BLOCK-CUSTOMER-V4, causing the fault. Its parent uses EXPORT-BGP. Delete the override to inherit that policy.

```
gnmic -a edge01.bob1.reefnet.test:57400 \
  -u admin -p 'NokiaSr11!' --skip-verify --encoding json_ietf \
  set --delete "$GROUP/export-policy"
gnmic -a edge01.bob1.reefnet.test:57400 \
  -u admin -p 'NokiaSr11!' --skip-verify --encoding json_ietf \
  get --type config --path "$GROUP"
```

Verify routes and service from both probes. Use `exit` to return to WSL. Change and rollback: `make solution SCENARIO=automation STEP=3`.

Measure and experiment

Watch traffic and loss

In WSL: choose one load. Wait at least 30 seconds at each level.

```
make traffic-40mbit
make traffic-60mbit
make traffic-75mbit
make traffic-10mbit
make traffic-diagnostics # inspect counters and receiver loss
```

Grafana: <http://localhost:3000>, folder **ReefNet / BOB1**. Check the interface, direction and time window.

In WSL: make enter `NODE=edge01.bob1.lagoontransit.test` In its router CLI:

```
info from state interface ethernet-1/1 statistics
```

Compare out-discarded-packets at two times.

Leave with quit. In WSL: make enter `NODE=service01.bob1.oceanresearch.test` Inside the service node:

```
tail -n 15 /tmp/iperf-server.log # received Mbit/s and loss
```

Control the experiment

In WSL: run `make traffic-10mbit`, then `make test`. Start only when all four results pass. Try one fault at a time.

Experiment	Inject	Restore
Link failure	<code>make fault-link</code>	<code>make clear-link</code>
Routing failure	<code>make fault-routing</code>	<code>make clear-routing</code>

Observe, restore, then repeat `make test` after convergence. Investigation: `make hint SCENARIO=monitoring STEP=4` (link), `STEP=5` (routing).

Short burst

```
make traffic-stop # stop continuous traffic first
```

For the rate-window comparison, wait five minutes. Then:

```
make traffic-burst-40mbit # stops after five seconds
make traffic-status # current load
```

Queries: `make hint SCENARIO=monitoring STEP=6`.

Stop and resume

Note your findings and next step, then stop with `make down`.

Scenario commands and `make reset` recreate the initial state. `make clean` also deletes lab state and NetBox data.